

Group 44 Local Screenshot Evidence

Information Retrieval Assignment 2

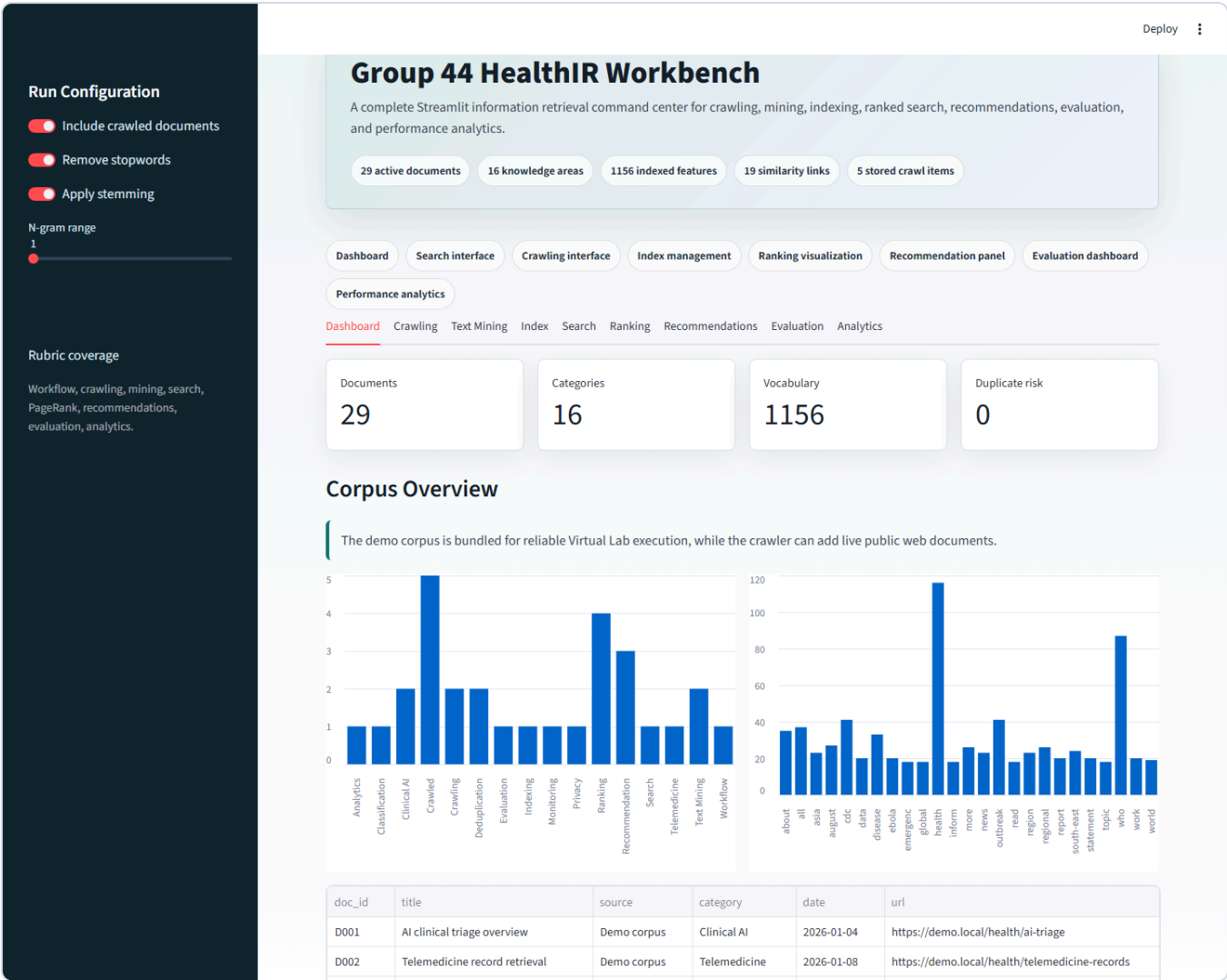
Application: Group 44 HealthIR Workbench

Local URL captured: `http://localhost:8501`

This report records local execution evidence for every Streamlit tab required by the assignment document: dashboard, crawling interface, text mining, index management, search, ranking visualization, recommendations, evaluation dashboard, and performance analytics.

Dashboard

Local Streamlit evidence for the Dashboard tab.



Crawling

Local Streamlit evidence for the Crawling tab.

Run Configuration

Include crawled documents

Remove stopwords

Apply stemming

N-gram range

1

Rubric coverage

Workflow, crawling, mining, search, PageRank, recommendations, evaluation, analytics.

Deploy

Group 44 HealthIR Workbench

A complete Streamlit information retrieval command center for crawling, mining, indexing, ranked search, recommendations, evaluation, and performance analytics.

29 active documents

16 knowledge areas

1156 indexed features

19 similarity links

5 stored crawl items

Dashboard

Search interface

Crawling interface

Index management

Ranking visualization

Recommendation panel

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Performance analytics

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Text Mining

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Search

Ranking

Recommendations

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Analytics

Configurable Web Crawling

Multiple seed URLs, depth control, URL normalization, and content hashing are used to manage acquisition and deduplication.

Seed URLs

https://www.who.int/news-room

https://www.cdc.gov

Crawling depth

1

Maximum pages

5

Stay on seed domain

Run crawler

Stored crawled documents

doc_id	title	source	url	date
C001	Newsroom	www.who.int	https://www.who.int/news-room	2026-08-22
C002	Centers for Disease Control and Prevention CDC	www.cdc.gov	https://www.cdc.gov/	2026-08-22
C003	World Health Organization (WHO)	www.who.int	https://www.who.int/	2026-08-22
C004	World Health Organization South-East Asia (WHO) - Home	www.who.int	https://www.who.int/southeastasia	2026-08-22
C005	WHO/Europe Home	www.who.int	https://www.who.int/europe	2026-08-22

Text Mining

Local Streamlit evidence for the Text Mining tab.

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Text Preprocessing and Mining

This screen turns raw text into structured features and exposes keyword extraction, document profiling, classification, and strategy comparison.

doc_id	title	category	tokens	unique_terms	top_terms
D001	AI clinical triage overview	Clinical AI	33	33	artificial, intelligence, clinical, triage, help
D002	Telemedicine record retrieval	Telemedicine	24	24	telemedicine, platform, generate, transcript, note
D003	Remote monitoring search	Monitoring	26	25	device, remote, patient, monitor, collect
D004	Clinical knowledge base indexing	Indexing	27	26	guideline, clinical, knowledge, base, store
D005	Duplicate discharge summaries	Deduplication	25	25	duplicate, clinical, document, often, appear
D006	Privacy in health retrieval	Privacy	26	25	inform, patient, privacy, regulation, require
D007	Biomedical query expansion	Search	24	23	expansion, biomedical, literature, search, query
D008	Hospital advisory crawling	Crawling	26	25	crawl, hospital, incident, dashboard, combine
D009	BM25 for medical search	Ranking	23	22	document, bm, rank, reward, contain
D010	TF-IDF vector search	Text Mining	25	25	tf-idf, vector, search, represent, document

Top corpus terms



Preprocessing strategy comparison

strategy	vocabulary_size	avg_doc_length	feature_density
raw tokens	1297	144.62	0.0629
stopword removal	1261	122.21	0.0592

Index

Local Streamlit evidence for the Index tab.

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Index Management

Metadata is stored in data/metadata.csv; document contents are stored separately in data/documents.csv.

Index build time

0.061 s

Average document length

122.21

Similarity graph edges

19

term

a-z

about

academy

acces

accessibility

accomplish

accountability

acros

act

action

Export active index metadata

Search

Local Streamlit evidence for the Search tab.

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Search Interface

Run TF-IDF, BM25, or Hybrid search with optional query expansion and PageRank authority blending.

Query

clinical search ranking

Retrieval model

Hybrid

Top K

8

Query expansion

PageRank authority weight

0.15

26 matching documents in 4.22 ms

1. HITS for health portals

HITS separates hubs and authorities in a linked collection. For health portals, a hub may point to many reliable guidelines, while an authority receives references from many trusted pages.

D012

Ranking

final 0.8940

text 0.9909

PageRank 0.0272

2. PageRank over similarity graphs

PageRank estimates importance through links or similarity relationships. In a document collection without hyperlinks, a similarity graph can approximate authority by connecting documents that cite similar concepts or share strong content overlap.

D011

Ranking

final 0.8823

text 0.9151

PageRank 0.0509

3. BM25 for medical search

Local Streamlit evidence for the Ranking tab.

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Ranking Visualization

PageRank is calculated on a document similarity graph to show how authority changes final ordering.

doc_id	PageRank	title
D007	0.0743	Biomedical query expansion
D022	0.0671	Public health web crawling
D008	0.0659	Hospital advisory crawling
C001	0.0613	Newsroom
C003	0.0611	World Health Organization (WHO)
D019	0.0595	Health document classification

Recommendations

Local Streamlit evidence for the Recommendations tab.

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Recommendation Panel

Top-K recommendations show similarity scores for content-based, collaborative-style, and hybrid modes.

Current document

D001 - AI clinical triage overview

Recommendation type

Hybrid

Top K

5

doc_id	title	category	similarity_score
D007	Biomedical query expansion	Search	0.3339
D022	Public health web crawling	Crawling	0.291
D008	Hospital advisory crawling	Crawling	0.2763
C003	World Health Organization (WHO)	Crawled	0.2717
C001	Newsroom	Crawled	0.2713

Evaluation

Local Streamlit evidence for the Evaluation tab.

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Evaluation Dashboard

Relevance judgements drive Precision, Recall, F1, Precision@K, Recall@K, AP/MAP, MRR, and NDCG comparisons.

Evaluation K

5

Evaluation authority weight

0.15

query	method	Precision	Recall	F1	Precision@5	Recall@5	AP	MRR	NDCG
clinical search ranking	TF-IDF	0.6	0.6	0.6	0.6	0.6	0.6	1	0.723
clinical search ranking	BM25	0.6	0.6	0.6	0.6	0.6	0.6	1	0.723
clinical search ranking	Hybrid	0.6	0.6	0.6	0.6	0.6	0.6	1	0.723
duplicate documents in corpus	TF-IDF	0.4	0.667	0.5	0.4	0.667	0.278	0.333	0.437
duplicate documents in corpus	BM25	0.6	1	0.75	0.6	1	0.533	0.5	0.68
duplicate documents in corpus	Hybrid	0.4	0.667	0.5	0.4	0.667	0.278	0.333	0.437
recommend similar medical documents	TF-IDF	0.6	0.75	0.667	0.6	0.75	0.65	1	0.788
recommend similar medical documents	BM25	0.6	0.75	0.667	0.6	0.75	0.65	1	0.788
recommend similar medical documents	Hybrid	0.6	0.75	0.667	0.6	0.75	0.65	1	0.788
evaluate retrieval system metrics	TF-IDF	0.4	0.667	0.5	0.4	0.667	0.467	1	0.651

Analytics

Local Streamlit evidence for the Analytics tab.

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Performance Analytics

Latency and index statistics make the system operationally inspectable, not just algorithmically correct.

query	method	latency_ms
clinical search ranking	TF-IDF	2.95
clinical search ranking	BM25	2.6
clinical search ranking	Hybrid	4.01
duplicate documents	TF-IDF	2.65
duplicate documents	BM25	2.32
duplicate documents	Hybrid	3.38
recommend similar medical documents	TF-IDF	2.39
recommend similar medical documents	BM25	3.46
recommend similar medical documents	Hybrid	4.83
web crawling hospital advisories	TF-IDF	3.6

4.0

3.5

3.0

2.5

2.0

1.5